ChatSafe: End-to-End Evaluation and Translation-Layer Ablation of a Translate-then-Classify Cyberbullying Detection Pipeline

John Lawrence S. Alovera
College of Computer and Information Science
Mapúa Malayan Colleges Mindanao
Davao City, Philippines
jlAlovera@mcm.edu.ph

Dianna Claire Marie C. Amihan
College of Computer and Information Science
Mapúa Malayan Colleges Mindanao
Davao City, Philippines
dcmAmihan@mcm.edu.ph

Izzy Mari C. Monteverde
College of Computer and Information Science
Mapúa Malayan Colleges Mindanao
Davao City, Philippines
imMonteverde@mcm.edu.ph

Rhodessa J. Cascaro
College of Computer and Information Science
Mapúa Malayan Colleges Mindanao
Davao City, Philippines
rjCascaro@mcm.edu.ph

Abstract*—Translation quality is often assumed rather than tested in translate-then-classify pipelines, even though downstream classification accuracy depends directly on how well the translation layer preserves meaning. This paper evaluates ChatSafe, a Translate-then-Classify pipeline for cyberbullying detection in code-mixed Filipino-Bisaya-English text, against a gold standard of 350 posts labeled by three licensed psychologists under blind majority vote. Three experiments are reported here, all using HateBERT, the classifier that the pipeline ultimately selects. A full-system evaluation against the gold standard comes first, followed by a controlled ablation that isolates what the translation layer contributes on its own. Rounding out the experiments is a three-way comparison of translation engines: a dedicated neural machine translation system (NLLB-200), a Southeast Asian regional language model (SEA-LION), and a general-purpose large language model (Gemini Flash), with each one assessed against a requirements framework covering slur and derogatory-term preservation, three-language code-switching, and Cebuano coverage. In deployment, the system reaches a Macro-F1 of 0.7044 and captures 81.75% of bullying cases—roughly 4 out of every 5 harassment instances in the dataset. Remove translation, and agreement with expert judgment falls from Cohen's κ = 0.4222 (Moderate) to κ = 0.1910 (Slight), suggesting the translation layer is structural to the pipeline rather than an incidental add-on. Derogatory intent survives in only one of the three engines tested, the general-purpose LLM. The dedicated NMT system strips it out through literal, token-level rendering. At the same time, the regional LLM loses it as well, in this case due to safety-driven sanitization exacerbated by gaps in its Cebuano coverage. What this suggests is that translation is doing domain alignment work here, not just bridging languages—and that preserving toxic register is its own requirement, separate from general translation quality. As Filipino-context language models mature in their Cebuano coverage, future iterations should test them as translation engines in their own right.

Keywords—translate-then-classify, machine translation, ablation study, low-resource NLP, cyberbullying detection, large language models, Filipino, Bisaya

I. Introduction

In a translate-then-classify pipeline, the translation layer sits upstream of every downstream decision. A slur neutralized in translation, a threat softened, an insult dropped during normalization: none of these can be recovered by a classifier trained afterward, since the signal never arrives. That makes the translation layer a single point of failure rather than a preprocessing convenience. Yet evaluations of such pipelines typically report only final classification metrics, leaving the contribution of the translation layer unmeasured.

ChatSafe is built on exactly this architecture for code-mixed Filipino-Bisaya-English cyberbullying detection, and its downstream classifier, HateBERT, a domain-adapted transformer pretrained on abusive English content, was fixed in an earlier benchmarking stage of this project for its lead on translated Filipino text. With the classifier fixed, three questions remain open: how the complete pipeline performs relative to expert judgment, how much of that performance the translation layer accounts for on its own, and whether the specific translation engine chosen changes the outcome.

This paper answers all three. First, the pipeline is evaluated end-to-end against a purpose-built gold standard labeled by three licensed psychologists, establishing system-level performance beyond the corpus-label benchmarks used earlier in this project. Second, a controlled ablation isolates the translation layer by comparing the pipeline with and without it under identical conditions. Third, three translation engines, a dedicated NMT system, a regional multilingual language model, and a general-purpose language model are compared against a requirements framework for preserving toxic and culturally embedded language, with an explanation of why two of the three fall short.

II. Related Work

A. Machine Translation for Cross-Lingual Hate Speech Detection

Machine-translated data has extended hate speech detection into new languages without native-language training sets. Casula and Tonelli study how annotation schemes and class imbalance interact with translated training data [1], and Pamungkas and Patti pair deep learning with multilingual lexicons for cross-domain and cross-lingual abuse detection [2]. Ranasinghe and Zampieri apply cross-lingual embeddings to the same identification task [3]. None of these isolates the translation step by controlled ablation; the present paper treats that isolation as a contribution in its own right.

B. Neural Machine Translation and Large Language Models for Low-Resource Text

NLLB-200 extends machine translation to over two hundred languages, several of them Philippine languages, but the breadth of coverage says nothing about whether register survives the crossing [4]. Subword schemes such as Byte-Pair Encoding [5] and SentencePiece [6] were built to keep rare and unseen words translatable by splitting them into smaller units. Still, the same splitting can strip meaning from words, slurs, and localized insults among them that depend on being read whole. Recent evaluations find that large language models translate general text about as well as dedicated NMT systems [7]; whether that parity extends to toxic or offensive register is a separate question this paper takes up directly.

C. Register and Toxicity Loss in Translation

Antypas and Camacho-Collados show that hate speech classifiers can lose more than 30 F1 points when evaluated outside their training distribution, a reminder of how little distortion a deployed system can tolerate before its output ceases to mean anything [8]. Zhao et al. find a related pattern one step earlier in the pipeline: large language models translating culturally loaded and offensive dialogue match human translators on general fluency but consistently tone down or neutralize the offensive terms themselves [9]. Safety training is a contributing factor. Röttger et al. document that models trained to be harmless through feedback learning will reject or sanitize prompts that only resemble unsafe ones, even when the prompt itself is benign [10], a tendency with obvious implications for a translation engine tasked with rendering toxic Filipino text into English without softening it.

D. Southeast Asian Language Models

SEA-LION is positioned as a language model built for the region [11], but its officially supported list runs to eleven languages, and Cebuano, the vehicle for Bisaya in this study's data, is not one of them. That gap is why SEA-LION appears below as a comparison point rather than a candidate to replace the pipeline's current translation engine.

III. Methodology

This work forms part of a larger thesis project developing the ChatSafe cyberbullying detection system; this paper focuses specifically on end-to-end evaluation of the deployed pipeline, the measured contribution of its translation layer, and a comparison of candidate translation engines. ChatSafe is a Translate-then-Classify pipeline for detecting cyberbullying in code-mixed Filipino-Bisaya-English social media text. Raw text is preprocessed, translated into English through a large language model, and classified by a transformer model into a binary Bullying or Not Bullying label with an accompanying confidence score.

A. Pipeline and Selected Classifier

ChatSafe processes each post through five stages before a label is produced. Raw text is first assigned a persistent identifier and deduplicated, then passed through a shield-and-restore preprocessing stage: hashtags, emoji, personally identifiable information such as usernames, leetspeak substitutions, censored profanity, typographic emphasis, and non-standard Unicode are each masked behind stable placeholder tokens and logged against the post's identifier, so that translation cannot destroy features the classifier will later need to see. The shielded text then passes to Gemini Flash, prompted for literal, register-preserving translation into English rather than free paraphrase. Once translation returns, the fourth stage reverses the shielding, restoring the original hashtags and emoji and normalizing punctuation, whitespace, and case to match the format the classifier was trained on. The fifth and final stage is classification itself: HateBERT, a BERT variant retrained on roughly 1.4 million banned Reddit comments [12], reads the restored English text and outputs a binary label (Bullying or Not Bullying) with a confidence score. HateBERT was chosen as the pipeline's classifier in an earlier benchmarking stage of this project for reaching the highest Macro-F1 on translated Filipino text among five architectures tested (0.6780) and the smallest cross-lingual performance drop of the group, at 9.2 percent. Fig. 1 shows the resulting architecture, scoped to this deployment path rather than the full multi-model benchmarking setup described elsewhere.

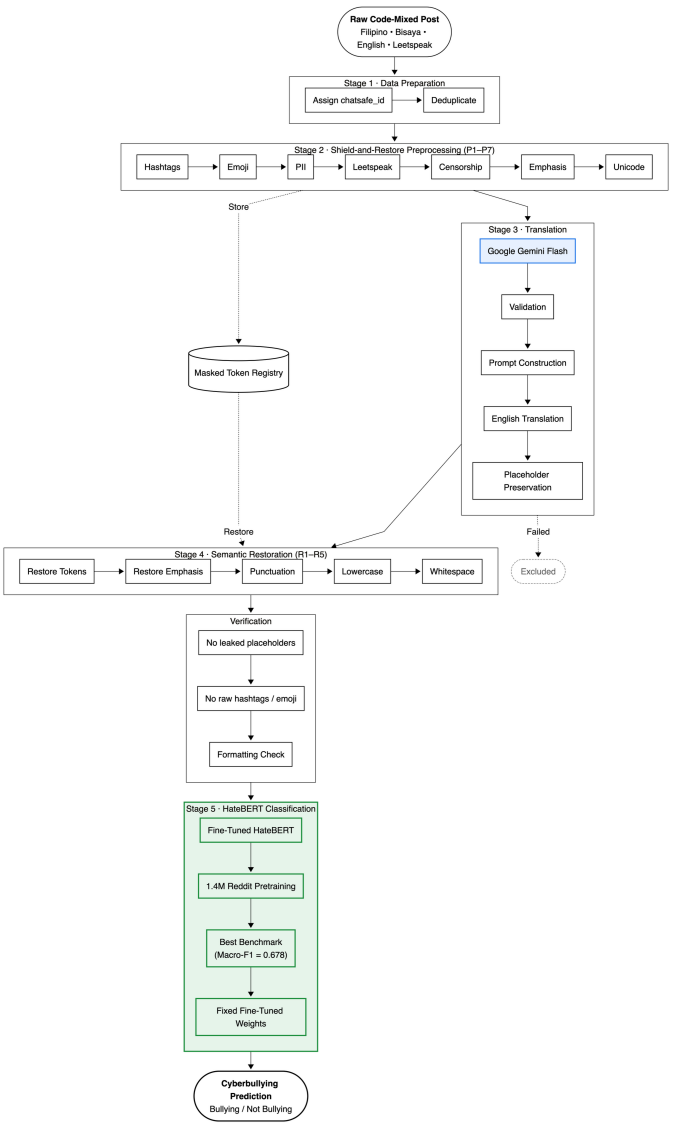


Fig. 1. ChatSafe deployed pipeline architecture.

B. Gold Standard Evaluation Set

The evaluation set used throughout this paper consists of 350 posts labeled by three licensed psychologists working independently and blind to each other's decisions, with the final label set by majority vote: 137 posts as Bullying and 213 as Not Bullying. This gold standard, rather than the corpus-derived labels used in earlier benchmarking, is what the results below are measured against.

C. Reliability Metrics

HateBERT's predictions on the gold standard are scored two ways: against the expert majority-vote label using standard classification metrics, and separately against the same label using a chance-corrected agreement statistic reserved for comparison with human judgment. Let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives. Performance is reported using accuracy (1), precision (2), and recall (3),

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

and Macro-F1 (4), the unweighted mean of the per-class F1 score across the C = 2 classes,

$$Macro{-}F_1 = \frac{1}{C} \sum_{c=1}^{C} F1_c \quad (4)$$

Macro-F1 is treated as the primary metric rather than accuracy because the gold standard is imbalanced, 137 Bullying against 213 Not Bullying, and accuracy alone would favor the majority class.

Agreement between HateBERT's predictions and the expert majority-vote label is measured separately from these four, using Cohen's κ:

$$\kappa = \frac{P_0 - P_e}{1 - P_e} \quad (5)$$

where P_0 is the observed agreement between the pipeline and the expert label, and P_e is the agreement expected by chance alone. Unlike raw accuracy, κ corrects for the agreement two labelers would reach by guessing, which matters here because the gold standard's roughly 60/40 class split would otherwise inflate agreement figures on its own. κ is interpreted on the Landis and Koch scale [13], [14], reproduced in Table I.

TABLE I. LANDIS AND KOCH INTERPRETATION BANDS

κ Range	Interpretation
< 0.00	Poor
0.00 – 0.20	Slight
0.21 – 0.40	Fair
0.41 – 0.60	Moderate
0.61 – 0.80	Substantial
0.81 – 1.00	Almost Perfect

D. Ablation Design

To isolate the translation layer's contribution, HateBERT was run twice on the gold standard under otherwise identical conditions: once on raw preprocessed Filipino text, and once on the same text after translation. Macro-F1 and Cohen's κ are compared against the expert labels across the two conditions; any difference is attributable solely to translation. This is a different ablation from the one reported in the companion benchmarking study, which measures all five architectures against the larger, corpus-labeled Filipino evaluation set. The ablation reported here measures only the deployed classifier, and specifically against expert psychological judgment rather than a repurposed proxy label. Hence, the quantity of interest aligns with human judgment rather than a corpus label.

E. Translation-Engine Comparison Design

Three engines were compared against a three-part requirements framework: whether culturally embedded derogatory and slur terms survive translation intact, whether three-language code-switching (Filipino, Bisaya, English) is handled without breaking down, and whether Cebuano vocabulary is specifically covered. NLLB-200 represents a dedicated, non-LLM neural machine translation system; SEA-LION represents a regional large language model tuned for Southeast Asian languages; Gemini Flash represents a general-purpose large language model with no specific regional tuning. Table V reports how each engine performs against the framework, illustrated with representative posts drawn from the gold standard.

IV. RESULTS

A. End-to-End Pipeline Performance

Table II reports the full performance profile of the deployed system on the gold standard. HateBERT achieves a Macro-F1 of 0.7044 and an accuracy of 70.57%, with a bullying recall of 81.75% (112 of 137 actual bullying posts correctly flagged), and a precision of 0.7166. Agreement with the expert majority label yields Cohen's κ = 0.4222, which is Moderate on the scale defined in Section III-C [13], [14]. Fig. 2 presents the confusion matrix underlying these figures: 78 false positives against 25 false negatives (Table III), a ratio that favors catching real harm over avoiding false alarms.

TABLE II. END-TO-END SYSTEM PERFORMANCE ON GOLD STANDARD (N=350)

Metric	Value
Accuracy	70.57%
Precision	0.7166
Recall (Bullying)	81.75%
Bullying F1	0.6850
Not Bullying F1	0.7239
Macro-F1	0.7044
Cohen's κ	0.4222 (Moderate)

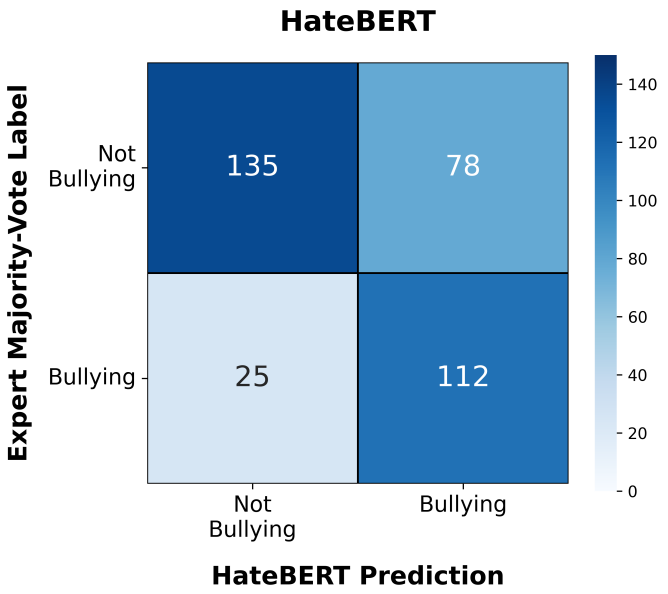


Fig. 2. Confusion matrix, HateBERT predictions vs. expert majority-vote label.

TABLE III. HATEBERT CONFUSION MATRIX, GOLD STANDARD (N=350)

Metric	Pred. Not Bullying	Pred. Bullying
Actual Not Bullying (n=213)	135 (TN)	78 (FP)
Actual Bullying (n=137)	25 (FN)	112 (TP)

B. *Ablation: Contribution of the Translation Layer*

Table IV compares HateBERT under the two ablation conditions. Removing translation and classifying preprocessed Filipino text directly drops Macro-F1 to 0.5879 and collapses agreement with the expert labels to $\kappa = 0.1910$, Slight on the same scale [13], [14]; every other metric falls in step, bullying-class F1 alone drops from 0.6850 to 0.4454. Restoring translation recovers all of it and more: a gain of 0.1165 in Macro-F1 and 0.2312 in κ. Fig. 3 plots this recovery across all six metric dimensions. The scale of the collapse, not just its direction, is the finding: without translation, the deployed classifier is barely better than chance agreement with trained human judgment.

TABLE IV. ABLATION: WITH VS. WITHOUT TRANSLATION (HATEBERT, N=350)

Metric	Without Translation	With Translation	Gain
Macro-F1	0.5879	0.7044	+0.1165
Bullying F1	0.4454	0.6850	+0.2396
Accuracy	0.6371	0.7057	+0.0686
Precision	0.6105	0.7166	+0.1061
Recall	0.5899	0.7257	+0.1358
Cohen's κ	0.1910 (Slight)	0.4222 (Moderate)	+0.2312

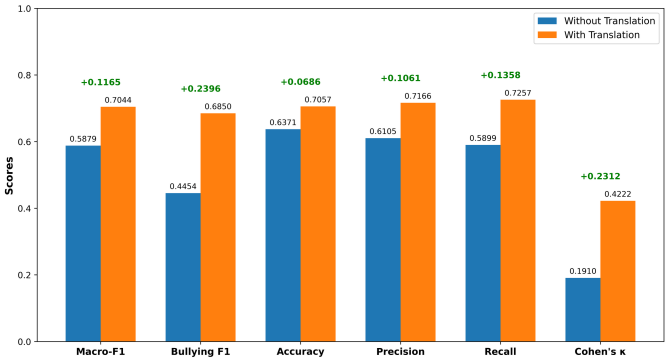


Fig. 3. Grouped bar chart, ablation across six metrics.

C. *Translation-Engine Comparison*

Table V scores the three engines against the requirements framework. NLLB-200 nominally satisfies one of the three requirements: Cebuano is among its 200-plus officially listed languages [4], though listed coverage does not translate into functional competence. Its subword vocabulary still fragments unfamiliar Filipino and Bisaya terms into pieces that carry no toxic signal, a token-bleeding failure consistent with how BPE and SentencePiece are known to behave on rare words [5], [6]. In the specific cases examined here, this failure surfaced as outright hallucination rather than simple neutralization, and the collapse extended even to a language NLLB formally supports. SEA-LION handles Tagalog-English switching adequately but lacks official Cebuano support and exhibits a second, independent failure mode: outputs are frequently softened relative to the source, consistent with what over-sanitization safety-tuned models exhibit on borderline content [10]. Gemini Flash is the only engine to satisfy all three requirements, preserving derogatory intent through what functions as contextual pragmatic inference rather than literal substitution. The derogatory-term-preservation column of Table V is illustrative rather than exhaustive: it reflects two culturally loaded terms, examined in detail below, chosen for their combination of low-resource specificity and high potential for information loss in translation, rather than scores averaged over a larger held-out sample. The code-switching and Cebuano-coverage columns, by contrast, rest on each engine's documented language support and switching behavior rather than on these two cases. The derogatory-term pattern nonetheless aligns with independent, peer-reviewed evidence from a different code-mixed language pair: a recent evaluation of machine-translated hate speech detection for English-Tamil content found translation measurably degraded classification performance specifically because slang and culturally loaded terms were mistranslated into semantically plausible but harmless output, the same failure mode documented here for Filipino and Bisaya [15].

The Tagalog derogatory term *palamunin* concretely illustrates the gap. Rooted in *kain* ("to eat"), it denotes an able-bodied dependent who contributes nothing to a household; it is not a slur in the identity-based sense but carries a strong insult register in everyday use. Fig. 4 traces the word through the full pipeline for all three engines, from raw input through tokenization, model input, generation, and final output.

TABLE V. TRANSLATION ENGINE REQUIREMENTS FRAMEWORK

Engine	Derogatory-Term Preservation	3-Language Code-Switching	Cebuano Coverage
NLLB-200 (dedicated NMT)	Fails — token fragmentation	Partial	Supported
SEA-LION (regional LLM)	Fails — safety sanitization	Adequate (Tagalog-English)	Not supported
Gemini Flash (general LLM)	Preserved — pragmatic inference	Handles all three languages	Covered functionally

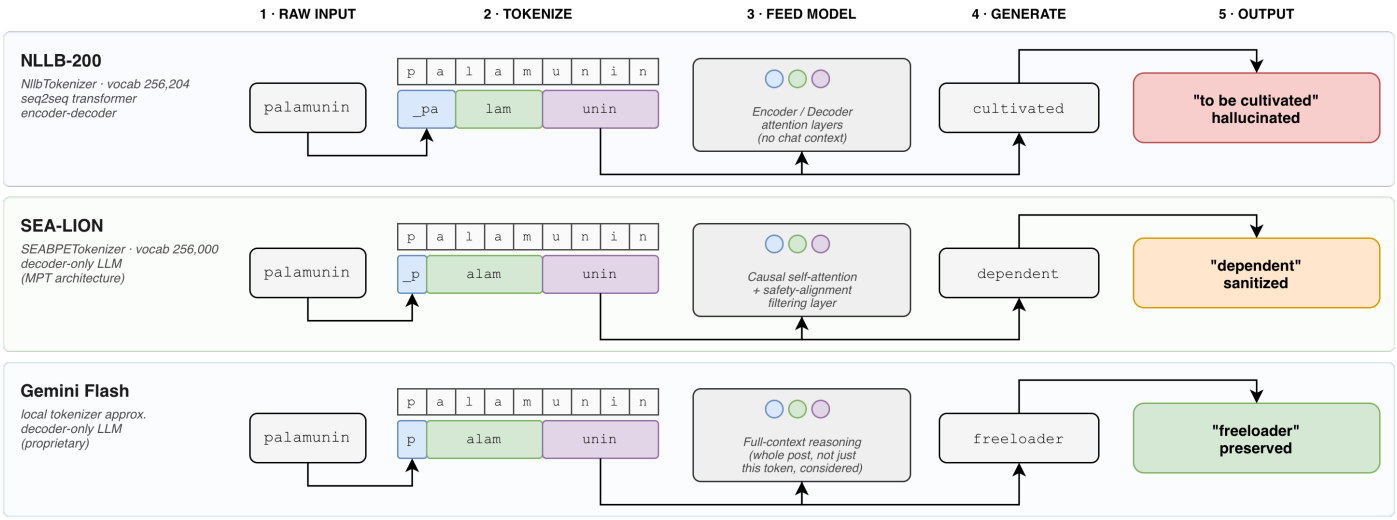


Fig. 4. Pipeline trace of the derogatory term "palamunin" across three translation engines, from tokenization through final output

Tokenization is nearly identical across engines; the divergence emerges only at generation. NLLB-200 fragments the word and produces the unrelated "cultivated," SEA-LION sanitizes it to the neutral "dependent," and Gemini Flash alone resolves it to "freeloader," preserving the term's contempt. A second example, the political portmanteau *kakampuke* (*kakampink*, a supporter of a 2022 presidential candidate, blended with a vulgar term), shows the same pattern: NLLB-200 again hallucinates an unrelated word ("thickness"), SEA-LION neutralizes it to "supporters," and Gemini Flash preserves its toxic English equivalent.

V. DISCUSSION

The ablation and the end-to-end result point to the same mechanism. Translation does not simply grant the classifier linguistic access it lacked; it moves the input closer to the distribution from which HateBERT was pretrained: colloquial, hostile English-language text drawn from banned Reddit comments [12]. Remove that step, and the classifier is reading Filipino text it was never built to parse, hence the collapse to near-chance agreement. This corresponds to an earlier finding in this project's benchmarking work: translation converts a cross-lingual problem into a cross-domain one. It extends it from corpus-level metrics to a single deployed model tested against expert judgment.

The engine comparison adds a second, separable point. General translation competence and toxic-register preservation are not the same capability, and evaluations that measure only the former risk certifying an engine that quietly fails the latter. NLLB-200 and SEA-LION are each competent systems for their intended purposes; neither was built to keep a slur intact along the way. Gemini Flash was not selected here for being a superior general translator; nothing in this comparison establishes that. It was selected because it happened to preserve the register that this specific downstream task needs. A single point of failure remains upstream of every metric in Table II. If a future translation engine sanitizes as SEA-LION does, the classifier below it never gets the chance to be wrong or right, since the harmful content it needs to see is already gone.

Characterizing why the pipeline disagrees with experts on specific posts is a deliberately separate research question from measuring how much it agrees in aggregate, which is this paper's scope; the 103 posts on which HateBERT's predictions disagree with the expert label are accordingly not analyzed pattern by pattern here. A companion validation study examines disagreement directly, decomposing it into inter-rater agreement and characterizing the error taxonomy that emerges when translation and classification are both accounted for. What this paper shows is that the pipeline operates at a Moderate level of agreement with trained human judgment, Cohen's $\kappa = 0.4222$, and that removing translation collapses this to Slight; what specifically separates the cases where the pipeline agrees with experts from the cases where it does not is the companion study's contribution, not this one's.

Two scope limitations qualify these findings. The gold standard evaluation set comprises 350 posts, a size dictated by the cost of blind, independent labeling by three licensed psychologists rather than an arbitrary convenience threshold; a larger held-out set would tighten the confidence around each point estimate in Table II, but is unlikely to change the

qualitative pattern, since the with/without-translation gap in Table IV is large relative to what sampling noise at this scale could plausibly produce. Second, every result in this paper concerns HateBERT specifically, the single classifier this project's earlier benchmarking stage selected; the ablation demonstrates that translation matters for this deployed model, not that its magnitude would be identical for a different architecture, and the companion benchmarking study's five-model, corpus-labeled ablation should be consulted for cross-architecture patterns rather than assuming this paper's exact coefficients generalize beyond HateBERT.

VI. Conclusion

This paper evaluated the ChatSafe pipeline end-to-end against expert-validated ground truth, finding a Macro-F1 of 0.7044 and a bullying recall of 81.75 percent on a 350-post gold standard. A controlled ablation showed that this performance is not incidental to the translation layer: removing it collapses agreement with expert judgment from Moderate ($\kappa = 0.4222$) to Slight ($\kappa = 0.1910$). A three-way comparison of translation engines further showed that this contribution is not interchangeable across translators; a dedicated NMT system and a regional language model both lose the toxic register that a general-purpose large language model preserves, for different underlying reasons: subword fragmentation in one case and safety-driven sanitization in the other. Translation, across both experiments, functions as domain alignment rather than a language bridge, and preserving derogatory intent is a distinct requirement from translation quality generally. As Filipino-context language models mature and extend their Cebuano coverage, they warrant evaluation as translation engines under this same framework, alongside continued fine-tuning of the classifier on expert-labeled data. A companion study extends this evaluation further, decomposing pipeline-expert disagreement into its component error patterns and the inter-rater ambiguity that the end-to-end numbers reported here raise but do not, on their own, resolve.

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